

OMAR GARRAOUI

Milan, Italy

ogarraoui@icloud.com — omargarraoui.com — github.com/omargarraoui

EDUCATION

B.Sc. Mathematical Engineering (Applied Mathematics), Politecnico di Milano Sep 2025 – Jun 2028

High School Diploma, ITST J.F. Kennedy Sep 2020 – Jun 2025

GPA: 3.9/4.0

EXPERIENCE

AI Researcher, Domy AI, Milan Jun 2026 – Present

- Work across the full **LLM training pipeline** – data preparation, **pre-training**, **fine-tuning**, **evaluation** – for a **400B+ parameter** frontier model on **large-scale distributed** infrastructure (Colosseum); specifics under NDA.

Research Associate, Polimi Data Scientists Dec 2025 – May 2026

- Designed and implemented **ML pipelines** on real industry datasets for **BCG**, **Bloomberg**, and **Amazon**; formulated hypotheses and tested them against real-world data.
- Presented technical findings directly to project sponsors, translating quantitative results into decision-relevant insight.

PROJECTS

Contradiction Tensor [\[GitHub\]](#)

Independent research project: scores every pair of beliefs for logical contradiction with a local **NLI cross-encoder** (DeBERTa-v3) into an $N \times N \times K$ tensor indexed by context, then computes the minimum belief set to drop for internal consistency via a **greedy 2-approximation of Minimum Vertex Cover** – exact revision is NP-hard.

CausalPulse [\[demo\]](#)

Independent research project: models the global news stream as a mixture of K unobserved **causal forces** and recovers them via **StrADiff** (arXiv:2604.04973), an unsupervised structural attribution method; surfaces which upstream force drives apparently unrelated coverage, where standard classification sees only surface similarity.

Prediction Market Arbitrage Scanner [\[GitHub\]](#)

Formulated and tested a pricing model for risk-free arbitrage across prediction market venues under multiple fee regimes ($c = \text{ask}(1 + r)$ taker, $c = \text{ask}/(1 - (1 - \text{ask})r)$ profit-deferred); shipped as a **Python/FastAPI** service running continuously in production on live market data.

HONORS & COMPETITIONS

ITACPC (Italian Collegiate Programming Contest) – 3rd place nationally, competing solo against 3-person teams in a timed algorithms and data structures contest.

Italian Mathematics Olympiad (2024) – National Semifinals qualifier, team captain.

Collegio di Milano Fellowship (2025, 2026) – merit-based, full scholarship.

PUBLICATIONS

Optimizing Exact String Matching via Statistical Anchoring [\[arXiv:2601.03271\]](#) – Enhances Boyer-Moore-Horspool by anchoring first mismatch checks to $c^* = \arg \min_{c \in P} \Pr(c \mid \text{corpus})$; faster window rejection at no added shift-heuristic complexity.

SKILLS

Programming	Python, C, C++, Java, SQL, FastAPI, SQLite, Git, $\text{\LaTeX}$
ML & Data	PyTorch, Transformers, scikit-learn, NumPy, Pandas, Excel
Languages	English (C2), Arabic (native), Italian (native), French (B2)